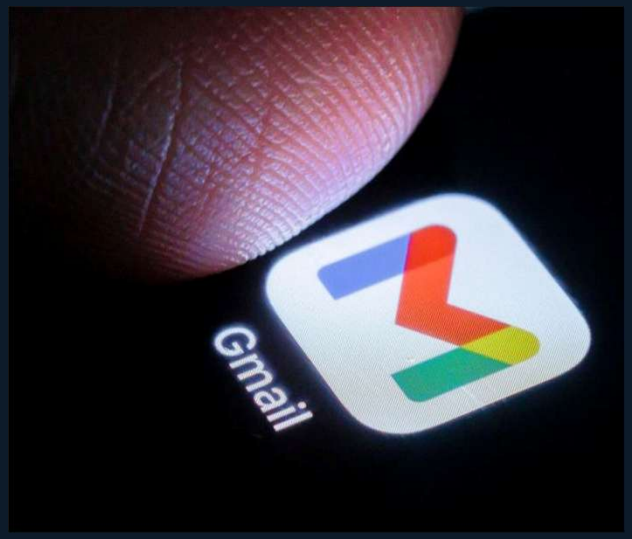


PROBLEM

STATEMENT

AI EMAIL TRIAGE SYSTEM

Solving the Email Overload Crisis with AI

OpenEnv-Compliant Reinforcement Learning Environment

THE PROBLEM



Email Usage Impact

2.5

hours / day

Average time spent on email

Email overload is a universal, costly problem.

Professionals across every industry are drowning in emails. Critical messages get buried under spam and low-priority notifications. Manually triaging hundreds of emails daily is:

- Time-consuming — 2.5 hrs/day = 30% of an 8-hr workday
- Mentally exhausting — decision fatigue from repetitive choices
- ⚠ Error-prone — important emails missed; spam slips through
- Expensive — lost productivity costs billions annually

WHY MANUAL TRIAGE FAILS

No Scalability

As email volume grows, manual effort scales linearly. A person can only read so fast.

Inconsistent Decisions

Humans classify email differently based on mood, fatigue, and context — no reliable standard.

Delayed Response

Important emails sit unread for hours. SLAs are missed. Stakeholders are frustrated.

No Learning Loop

Humans don't automatically improve at triage over time without explicit training programs.

Misclassification

Spam filters are too aggressive (false positives) or too lenient (false negatives).

Decision Fatigue

The 50th email of the day gets far less cognitive attention than the 1st one.

CURRENT STATE vs IDEAL STATE

✗ CURRENT STATE	☑ IDEAL STATE
✗ Humans read every single email manually	✓ AI auto-classifies: spam / important / normal
✗ No consistent classification criteria	✓ Consistent, rule-based decision framework
✗ Priority decisions vary by person	✓ Priority auto-assigned: high / medium / low
✗ Replies written from scratch each time	✓ Smart reply suggestions generated instantly
✗ 2.5 hrs/day wasted on email management	✓ Time saved redirected to high-value work
✗ No feedback loop for continuous improvement	✓ Reinforcement learning improves over time

THE SCALE OF THE PROBLEM

Global email overload by the numbers

347B

Emails sent & received every day globally

2.5h

Average daily time spent on email

$650B

Annual productivity loss from email overload

85%

Workers feel overloaded by email volume

The Opportunity

AI-powered email triage can reduce time spent on email by up to 70%, giving back 1.5–2 hours of focused work time every single day. At enterprise scale, this translates to millions in recovered productivity.

THREE CORE CHALLENGES

01



Classification

CHALLENGE: Which emails are spam, important, or just normal?

Spam filters miss nuanced phishing. Newsletters get confused with important work emails. Borderline cases require context and judgment that simple filters lack.

02



Prioritisation

CHALLENGE: Which emails need attention NOW vs. can wait?

Not all important emails are equally urgent. A server outage needs immediate response; a project update can wait. Humans must judge urgency from context — a cognitively expensive task.

03



Reply Generation

CHALLENGE: What's the right professional response?

Composing appropriate, context-aware replies takes significant time. Each reply requires reading the email fully, understanding context, and crafting professional language — multiplied by hundreds of emails.

THE SOLUTION EXISTS

Reinforcement Learning + AI Agent

An AI agent trained using step-by-step reinforcement learning signals can automatically triage your inbox — classifying, prioritising, and replying — with accuracy that improves over time.



Classify emails instantly



Prioritise with context-awareness



Generate professional replies

THANK YOU